

A Data sources

ARCHS4 mouse

997,515 profiles audited
17,244 selected across 20 tissues
Complete GEO-series splits

NASA OSDR API

1,610 biological profiles
75 accessions; FLT/GC labels
Study and material retained

Biological scope

Pooled multi-tissue generation
Tissue-specific analysis
Skeletal-muscle groups retained

B Configurable generative benchmark

Expression

Raw, CPM, TPM
log transforms
z-score, robust, MaxAbs

Harmonization

None; two study z-scores
ComBat/ComBat-seq
MBatch; MOBER

Generator

WGAN-GP; DDIM
GeneJEPa screened as
representation only

Training scope

OSDR only; ARCHS4 only
ARCHS4 pretrain plus
OSDR adaptation

Cohort structure

One or many studies
pooled or per tissue
accession balancing

Conditioning

FLT/GC; tissue; study
material; muscle group
and available covariates

C Staged selection and biological use

Grouped splits

GEO series or OSDR
accessions kept intact

Independent gates

Fidelity, diversity,
memorization, effects

Selected DDIM

Only candidate passing
final joint locked gates

Synthetic-guided analysis

Per-tissue use selected;
real OSDR defines effects

The matrix defined a gated search, not an exhaustive Cartesian sweep. Synthetic profiles were never counted as additional animals.